

Name: _____

Roll No.: _____

Indian Institute of Information Technology Vadodara

Program: MCA

Semester: I (Autumn 2025)

Course ID: CS441

Course Name: Computer Programming and Problem Solving

Mid-Semester Exam

Marks: 30

Time: 120 Minutes

NOTE:

1. All questions carry equal marks.
2. No queries will be entertained in the exam hall; use your discretion while answering.
3. Write answers concisely and to the point. Avoid long paragraphs (use bullet points wherever suitable).

Questions

Q1. A)

[2 Marks]

Write an algorithm to check the body temperature of a person based on a thermometer reading.

- Formula: Temperature is measured in °C.
- Person should be **normal** if temperature is between 36.5°C and 37.5°C.
- Person should have **fever** if temperature is greater than 37.5°C.
- Person should be **low temperature** if temperature is less than 36.5°C.

B)

[3 Marks]

Write a C program to calculate the water bill of a household according to the given conditions.

- For first 1000 liters: Rs. 2.00 per 100 liters
- For next 2000 liters: Rs. 3.00 per 100 liters
- For next 2000 liters: Rs. 5.00 per 100 liters
- For usage above 5000 liters: Rs. 8.00 per 100 liters
- An additional service charge of 10% is added to the bill

Q2. (a) What is the use of qualifiers or type modifiers in C? Explain with an example.

[2 Marks]

(b). What is the difference between while and for loops? Write a C program to compute the binomial coefficient.

[3 Marks]

$${}^nC_r = \frac{n!}{r!(n-r)!}$$

Q3. Write a C program to read marks for a minimum of 6 subjects and a maximum of 10 subjects (use a loop for input). Calculate the total marks and average. Identify the highest and lowest subject marks. Assign a grade based on the average with the following grading scheme:

- o Grade A: 90 and above
- o Grade B: 75–89
- o Grade C: 60–74
- o Grade D: 45–59
- o Grade P: 35–44
- o Grade F: below 35

[5 Marks]

If the student fails in **any one subject (<35)**, the overall grade should be **F (Fail)**, regardless of the average.

Display the **total, average, highest mark, lowest mark**, and the **final grade**.

Q4. Consider the following code segments and find the output for each:

[4 Marks]

a). #include <stdio.h>

```
void main() {
    int x = 5;
    int y = x++ + ++x;
    printf("%d", y);
}
```

b). #include <stdio.h>

```
void main() {
    int a = 2, b = 3;
    int c = a-- * ++b;
    printf("%d", c);
}
```

c). #include <stdio.h>

```
void main() {
    int a = 10;
    int b = printf("%d", a--);
    printf("%d", b);
}
```

d). #include <stdio.h>

```
void main() {
    int x = 0;
    int y = printf("%d", x) && printf("%d", x+1);
    printf("%d", y);
}
```

Q5. Write a C program that accepts an integer as input and prints all numbers obtained by **cyclically rotating its digits to the left** by one position repeatedly until the original number is obtained again.

For example: Input: 123

Output:

231

312

123

Constraints:

(a) Do not use arrays or strings; use arithmetic operations only.

(b) The program should handle numbers of any length.

[5 Marks]

Q6. Write an **algorithm** and an **equivalent C program** that:

1. Reads a sentence (character sequence) from the user.

2. Counts and prints the number of: **Words**, **Digits**, **Special characters** (non-alphanumeric, non-space characters), and **Spaces**

Guidelines:

- Consider words to be sequences of characters separated by spaces.
- Use only loops and conditional statements ('if-else') for processing.
- Avoid using advanced string libraries if possible.

Example: Input: Hello World! 123

Output:

Words: 2

Digits: 3

Special characters: 1

Spaces: 2

[6 Marks]

--- Good Luck ---